

Bayesian Statistics: Techniques and Models

University of California, Santa Cruz

About this Course

This is the second of a two-course sequence introducing the fundamentals of Bayesian statistics. It builds on the course Bayesian Statistics: From Concept to Data Analysis, which introduces Bayesian methods through use of simple conjugate models. Real-world data often require more sophisticated models to reach realistic conclusions. This course aims to expand our “Bayesian toolbox” with more general models, and computational techniques to fit them. In particular, we will introduce Markov chain Monte Carlo (MCMC) methods, which allow sampling from posterior distributions that have no analytical solution. We will use the open-source, freely available software R (some experience is assumed, e.g., completing the previous course in R) and JAGS (no experience required). We will learn how to select, fit, assess, and compare Bayesian statistical models to answer scientific questions involving continuous, binary, and count data. This course combines lecture videos, computer demonstrations, readings, exercises, and discussion boards to create an active learning experience. The lectures provide some of the basic mathematical development, explanations of the statistical modeling process, and a few basic modeling techniques commonly used by statisticians. Computer demonstrations provide concrete, practical walkthroughs. Completion of this course will give you access to a wide range of Bayesian analytical tools, customizable to your data.



Taught by: Matthew Heiner, Doctoral Student
Applied Mathematics and Statistics

Level	Intermediate
Commitment	5 weeks of study, 4-6 hours/week.
Language	English Volunteer to translate subtitles for this course
How To Pass	Pass all graded assignments to complete the course.
User Ratings	★★★★☆ 4.9 stars

Syllabus

WEEK 1

Statistical modeling and Monte Carlo estimation

Statistical modeling, Bayesian modeling, Monte Carlo estimation

 11 videos, 4 readings [expand](#)

 **Graded:** Lesson 1

 **Graded:** Lesson 2

 **Graded:** Lesson 3

 **Graded:** Markov chains

WEEK 2

Markov chain Monte Carlo (MCMC)

Metropolis-Hastings, Gibbs sampling, assessing convergence

 11 videos, 7 readings [expand](#)

 **Graded:** Lesson 4

 **Graded:** Lesson 5

 **Graded:** Lesson 6

 **Graded:** MCMC

WEEK 3

Common statistical models

Linear regression, ANOVA, logistic regression, multiple factor ANOVA

 11 videos, 5 readings [expand](#)

 **Graded:** Lesson 7 Part A

 **Graded:** Lesson 7 Part B

 **Graded:** Lesson 8

 **Graded:** Lesson 9

 **Graded:** Common models and multiple factor ANOVA

WEEK 4

Count data and hierarchical modeling

Poisson regression, hierarchical modeling

 10 videos, 7 readings [expand](#) **Graded:** Lesson 10 **Graded:** Lesson 11 Part A **Graded:** Lesson 11 Part B **Graded:** Predictive distributions and mixture models**WEEK 5**

Capstone project

Peer-reviewed data analysis project

 1 video, 1 reading [expand](#) **Graded:** Data Analysis Project[View Less](#)

How It Works

GENERAL**How do I pass the course?**

To earn your Course Certificate, you'll need to earn a passing grade on each of the required assignments—these can be quizzes, peer-graded assignments, or programming assignments. Videos, readings, and practice exercises are there to help you prepare for the graded assignments.

✓ [More](#)

What do start dates and end dates mean?

Most courses have sessions that run multiple times a year — each with a specific start and end date. Once you enroll, you'll have access to all videos, readings, quizzes, and programming

assignments (if applicable). Peer-graded assignments can only be submitted and reviewed once per session has begun. If you choose to explore the course without purchasing you may not be able to access certain assignments. If you don't finish all graded assignments before the end of the session, your progress will be saved and you'll be able to pick up where you left off when the next session begins.

After you submit your work, you'll be asked to review your classmates' assignments. To pass, you'll

need to review a certain number of reviews.

How are grades calculated?

We suggest due dates to help you manage your schedule and keep your work from piling up. Quizzes and programming assignments can be submitted late without penalty. However, it will be possible to provide a score for each part of the assignment.

Final grades are calculated by combining the median scores you receive for each section three days of the assignment deadline.

What kind of feedback should I give?

Can I re-attempt an assignment?

Be encouraging and honest. Acknowledge what your classmate did well and offer constructive feedback. Your comments should reflect your learning and understanding of the material. The University of Melbourne, The Chinese University of Hong Kong, and the University of Malakaua are good examples of how to give feedback. It's important to review your work. In some cases you may need to wait before re-submitting a programming assignment or quiz. We encourage you to review course material during this delay.

Is there a penalty for submitting my work late?

No, it's important to submit your work as close to the due date as you can. Classmates grade most assignments within three days of the due date. If you submit yours too late, there may not be time to review your work.

If I fail an assignment, can I try again?

Yes, but you'll need to re-submit your work and any grade you've already received will be deleted. It's important to make sure you have enough time to grade your work.

Can I re-attempt an assignment?

Yes, but you'll need to re-submit your work and any grade you've already received will be deleted.



Advanced Linear Models for Data Science 2: Statistical Linear Models

Johns Hopkins University



Advanced Linear Models for Data Science 1: Least Squares

Johns Hopkins University

